The Monocyte Activation Test (MAT)

A human-relevant approach for pyrogen testing

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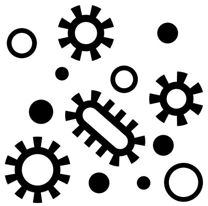
18th December 2025

What is pyrogenicity?

Pyrogen = fever-inducing substance

<u>Exogenous</u> pyrogens



From outside the body

- Microbial

Bacterial (e.g. ***endotoxins****), viral, fungal*

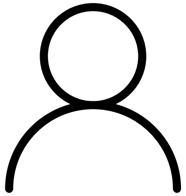
- Non-microbial

Particulate matter, chemicals, degradation products in drugs/devices

<u>Endogenous</u> pyrogens

Messengers of the immune system

Interleukin-6 (IL-6), interferon gamma (IFN-γ)…



Why is it important to test for pyrogens?

- **Risk of contamination**

  Pyrogens can be introduced during production or be intrinsically present.

- **Severe adverse effects**

  Fever, diarrhea, vomiting, and life-threatening endotoxic shock.

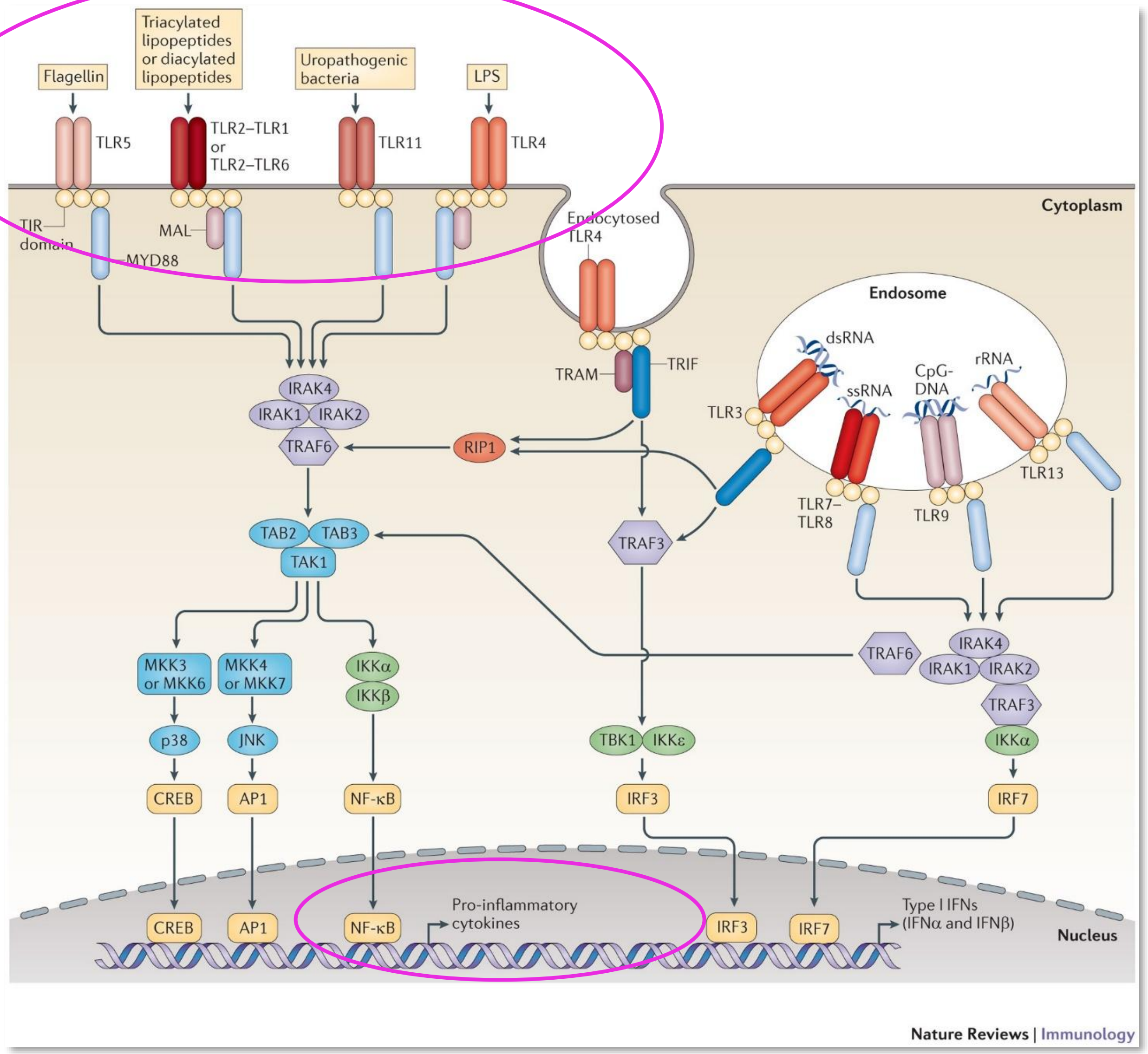
- **Critical for parenteral products**

  Bypass of the natural body barriers.

How to test for pyrogens?

	Bacterial endotoxin tests		Complete pyrogen tests	
Test type	Limulus Amoebocyte Lysate test (LAL)	Recombinant Factor C test (rFC)	Rabbit Pyrogen Test (RPT)	Monocyte Activation Test (MAT)
Detects endotoxins	✓	✓	✓	✓
Detects non-endotoxin pyrogens	✗	✗	✓	✓
Animal free	✗	✓	✗	✓
Regulatory compliance	Ph. Eur. 2.6.14 USP <85>	Ph. Eur. 2.6.32 USP <86>	Ph. Eur. 2.6.8 (pending suppression) USP <151>	Ph. Eur. 2.6.30
Mechanism	Clotting cascade in crab blood amoebocytes (*in vitro*)	Recombinant enzyme from first step LAL (*in vitro*)	Raise of body temperature after injection (*in vivo*)	Release of inflammatory cytokines from human monocytes (*in vitro*)

Monocyte Activation Test: principle



Activation of innate immune system by pyrogens

Toll-like receptors (TLR) recognize molecular patterns

Signalling cascade → production of cytokines

(O'Neill *et al.*, 2013)

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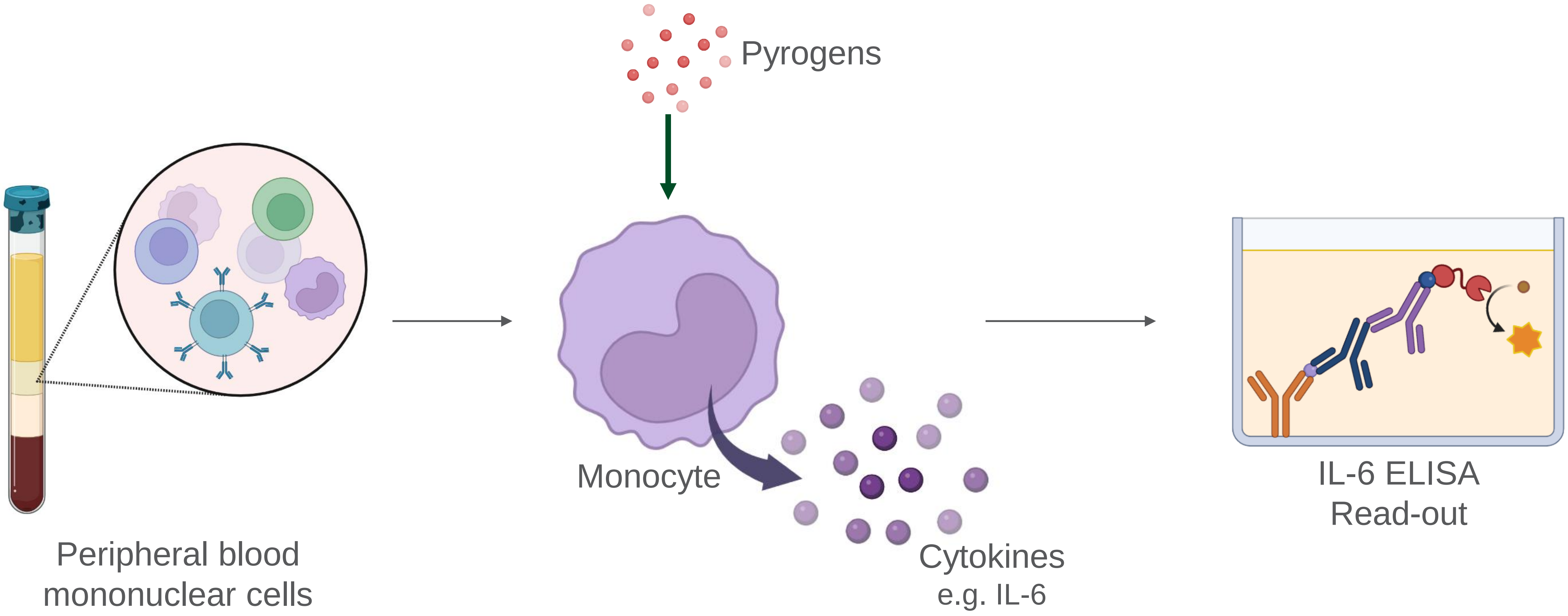
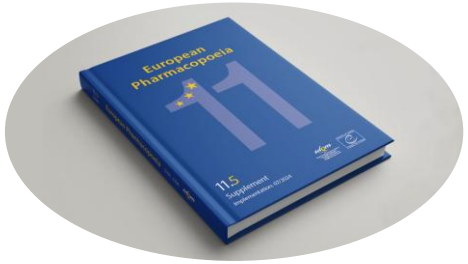
Monocyte Activation Test: principle



Other monocyte sources:
- Whole blood
- Continuous cell lines

Ph. Eur. 2.6.30



- **Method 1**: semi-quantitative test
    - 3 dilutions of test solution
    - 4 replicates
    - All dilutions spiked with LPS
    - Standard curve

- **Method 2**: reference lot comparison test
  For products that show interference
    - 3 dilutions of test solution
    - 3 dilutions of reference lot
    - Positive and negative controls

Table 2.6.30.-1

Solution	Solution/dilution factor	Added endotoxin	Number of replicates
A	Test solution/f	None	4
B	Test solution/f_1	None	4
C	Test solution/f_2	None	4
AS	Test solution/f	Equal to or near the middle of the endotoxin standard curve	4
BS	Test solution/f_1	Equal to or near the middle of the endotoxin standard curve	4
CS	Test solution/f_2	Equal to or near the middle of the endotoxin standard curve	4
R_0	Pyrogen-free saline or test diluent	None (negative control)	4
R_1-R_x	Standard endotoxin diluted in pyrogen-free saline or test diluent	≥ 4 concentrations of standard endotoxin	4 of each concentration

Table 2.6.30.-2

Solution	Solution/dilution factor	Number of replicates
A	Solution of reference lot/f_1	4
B	Solution of reference lot/f_2	4
C	Solution of reference lot/f_3	4
D	Solution of preparation to be examined/f_1	4
E	Solution of preparation to be examined/f_2	4
F	Solution of preparation to be examined/f_3	4
G	Positive control (standard endotoxin)	4
R_0	Diluent (negative control)	4

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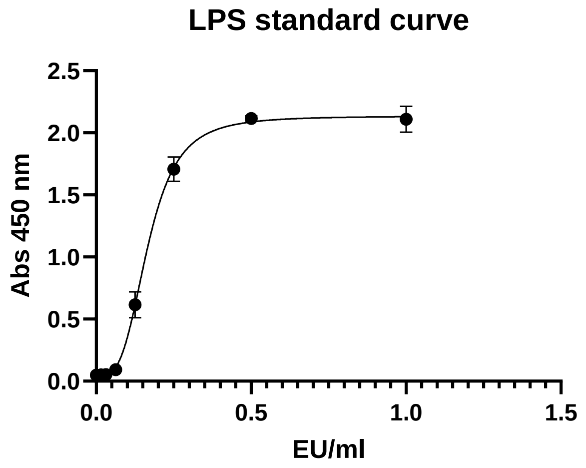
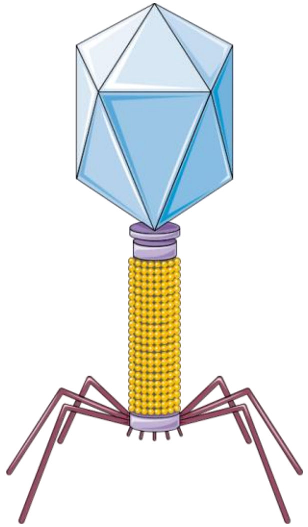
Criteria for validity of the test (Ph. Eur. 2.6.30)

- Qualification of cells
- Endotoxin standard curve
  - Appropriate dose-response curve:
    - Linear regression model (minimum 4 concentrations)
    - 4-/5-parameter logistic model (minimum 5/6 concentrations)
  - At least 4 replicates of each concentration
  - Goodness of fit
  - Coefficient of determination > 0,975
- The test detects endotoxins and non-endotoxin contaminants (positive controls & spikes)
- The test solution does not interfere with the test: LPS spike recovery between 50-200%
- The test solution does not interfere with the detection system (cytokine spike)



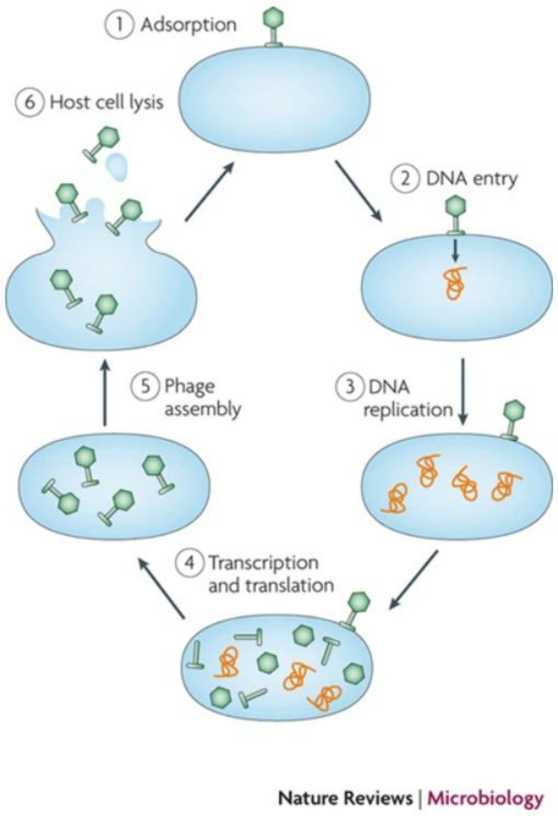
MAT application: phage medicinal products

Bacteriophage (phage)



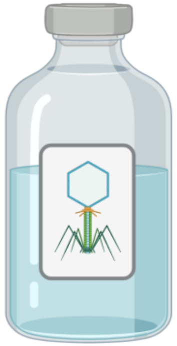
- Viruses of bacteria
- Ubiquitous in nature
- Highly specific

Phage life cycle



Labrie et al. (2010)

Phage therapy



Use of phages to treat bacterial infections, especially antibiotic-resistant strains

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MAT application: phage medicinal products

Host	Phage	MAT (EEU/ml)
P. aeruginosa	vB_Paer_1	< 17
	vB_Paer_2	< 3,125
	vB_Paer_3	< 6,25
	vB_Paer_4	< 3,125
	vB_Paer_5	15,893
	vB_Paer_6	< 3,125
	vB_Paer_7	< 3,125
	vB_Paer_8	< 6,25
A. xylosoxidans	vB_Axyx_1	190,74
K. pneumoniae	vB_Kpne_1	> 2500
	vB_Kpne_2	135,3
M. morganii	vB_Mmor_1	> 12500
S. aureus	vB_Sau_1	172,78
	vB_Sau_2	108
	vB_Sau_3	31,85
	vB_Sau_4	< 3,125
E. faecalis	vB_Efae_1	< 3,125
M. abscessus	vB_Mabs_1	< 3,125

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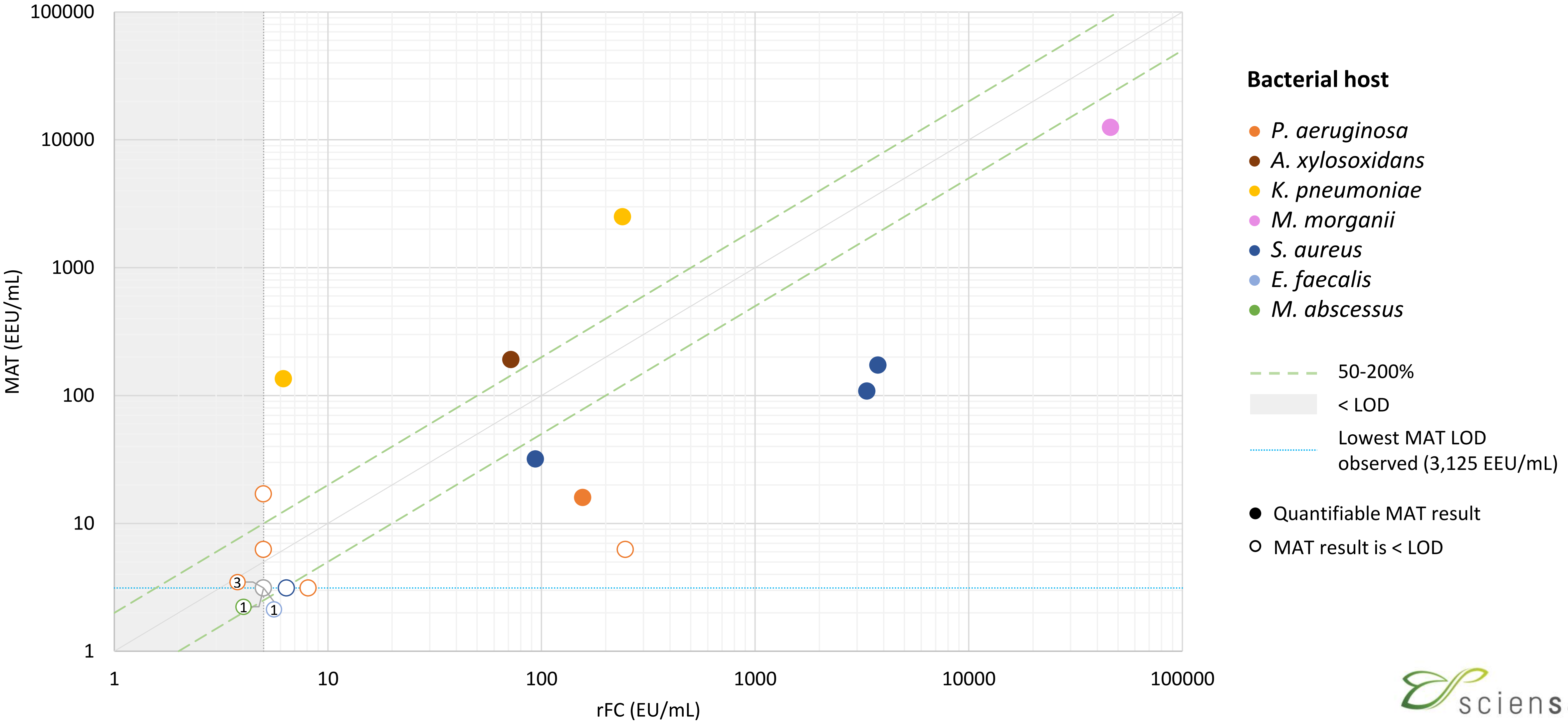
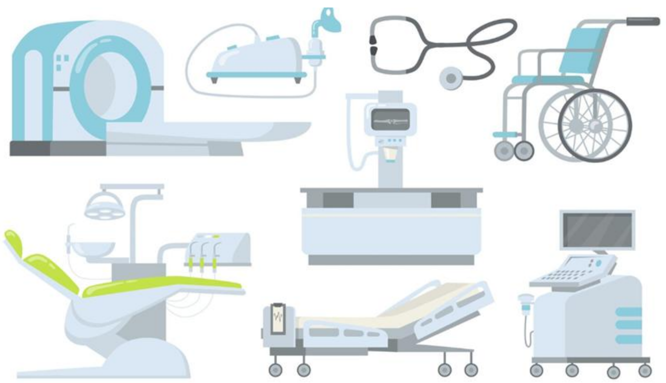
MAT vs. rFC: phage medicinal products

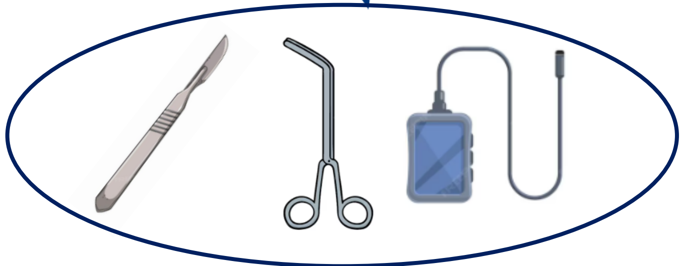


Application of the MAT to medical devices

Medical devices



Reusable



Decontamination life cycle

WHO, 2016

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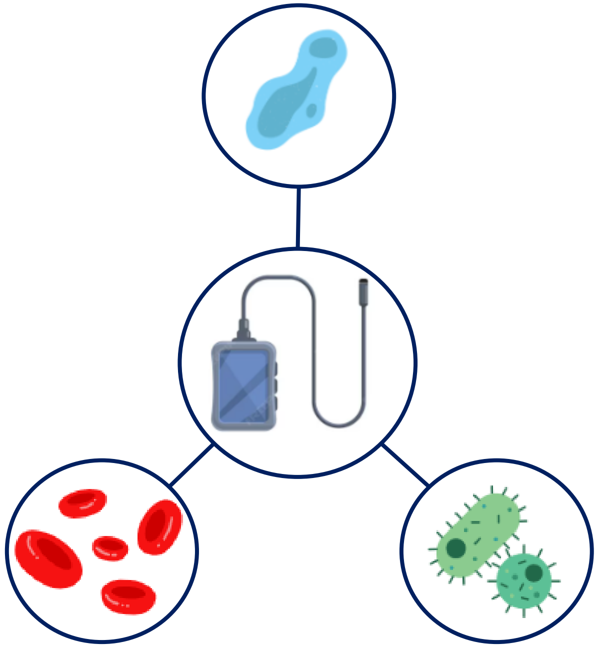
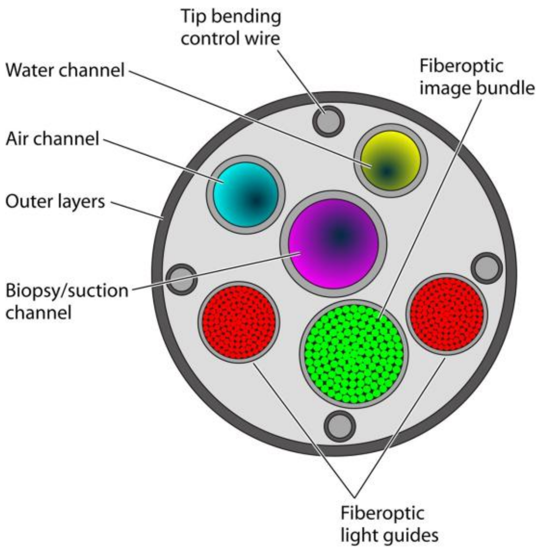
Application of the MAT to medical devices

Heavy contamination



Hard to clean



Kovaleva, 2013

Residual contamination

- Residual endotoxins and proteins persist after reprocessing
- Documented in studies across multiple medical devices

QMED Project

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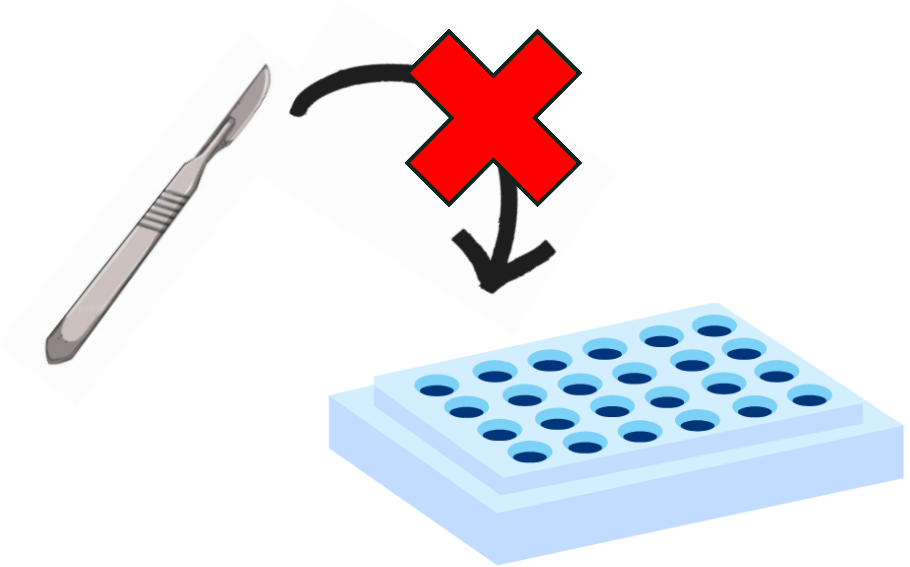
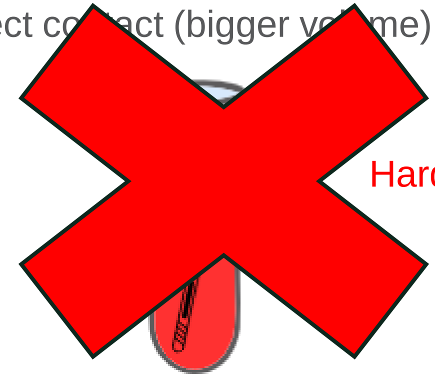
Application of the MAT to medical devices

- **Challenges linked to medical devices :**
  - Solid nature
  - Various shapes and sizes



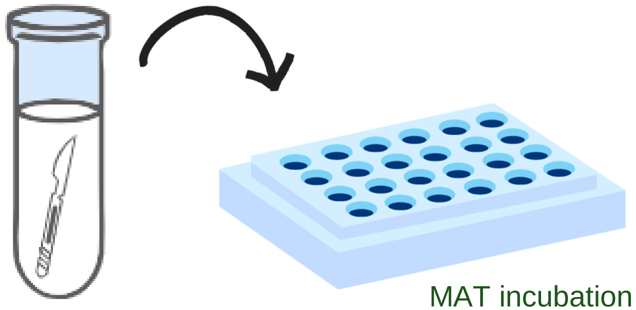
MAT incubation plate

- **Two possibilities:**
  - Direct contact (bigger volume)



Harder to standardize

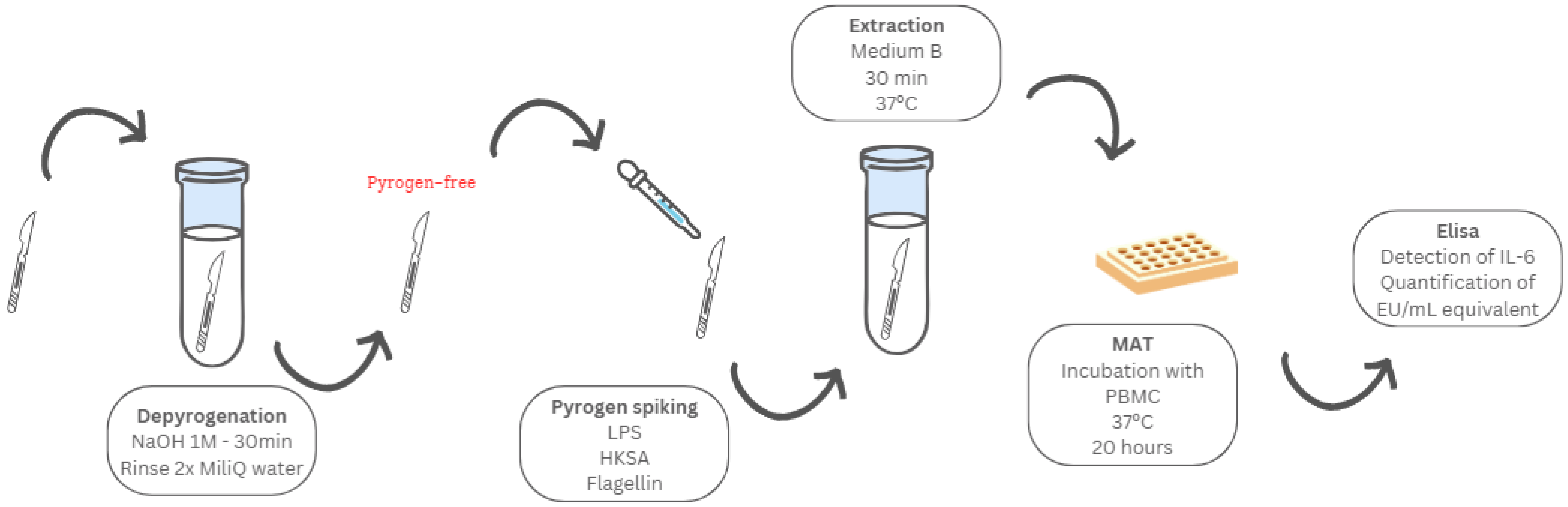
  - Indirect contact (extraction)



MAT incubation plate

Application of the MAT to medical devices

Final method



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Application of the MAT to medical devices

Results

LPS													
	OD	**OD**	**OD**	**OD**	**Average**	**Stdev**	**CV**	**[C]th**	**[C]**	**Recovery**	**Av Recovery**	**Stdev**	**CV**
13-12-24	1,562	1,392			1,4770	0,1202	8,1387	0,0625	0,0552	88,32	87,75	8,522809	9,7126
	1,535	1,638			1,5865	0,0728	4,5907	0,0625	0,0598	95,68			
19-02-25	0,622	0,444	0,43	0,36	0,4640	0,1116	24,0428	0,05	0,0379	75,8			
11-03-25	0,573	0,607	0,507		0,5623	0,0508	9,0420	0,05	0,0456	91,2			
HKSA													
	OD	**OD**	**OD**	**OD**	**Average**	**Stdev**	**CV**	**[C]Std**	**[C]**	**Recovery**	**Av Recovery**	**Stdev**	**CV**
19-02-25	0,223	0,215	0,256	0,208	0,2255	0,0212	9,4176	0,056	0,0287	51,25	51,8317	0,822671	1,5872
28-04-25	0,585	0,476	0,48		0,5137	0,0618	12,0329	39,7173	20,8172	52,41			
Flagelin													
	OD	**OD**	**OD**	**OD**	**Average**	**Stdev**	**CV**	**[C]Std**	**[C]**	**Recovery**	**Av Recovery**	**Stdev**	**CV**
19-02-25	0,354	0,387	0,334	0,434	0,3773	0,0437	11,5815	0,03	0,0345	115	116,3333	1,885618	1,6209
	0,406	0,365	0,354	0,466	0,3978	0,0507	12,7478	0,03	0,0353	117,67			

Table 4 - Quantification of pyrogenic contamination, expressed in endotoxin unit (EU) equivalents per milliliter, in standards and corresponding extracted samples. Samples were spiked with known concentrations of lipopolysaccharide (LPS), flagellin (FLG), and heat-killed Staphylococcus aureus (HKSA), then immersed in 10 mL of Medium B and incubated for 30 minutes at 37 °C. Supernatants were collected and subsequently incubated overnight with peripheral blood mononuclear cells (PBMCs), after which IL-6 concentrations were quantified by ELISA. OD: optical density, Stdev: standard deviation; CV: coefficient of variation (%); [C]th: theoretical concentration (EU/mL); [C]: real concentration (EU/mL); Av Recovery: average recovery.

Practical considerations

- **Challenges and limitations**:
    - Variability in cell sources (e.g., primary cells vs cell lines)
    - Need for standardized protocols and reference materials
    - Higher cost and complexity than LAL/rFC

- **Best practices for MAT application:**
    - Pyrogen-free consumables
    - Assessment of sample interference
    - Assessment of the maximum valid dilution

Future directions and conclusion

- **Future perspectives**:
  - Automation and high-throughput adaptations
  - Integration with other in vitro methods for comprehensive pyrogen testing
  - Potential expansion to new applications (e.g., nanomedicines, cell therapies)

- **Conclusions:**
  - Reliable and ethically favorable in vitro method
  - Detect a wider range of pyrogens
  - Human-relevant
  - Highly sensitive
  - Overall, effective and humane alternative to animal testing

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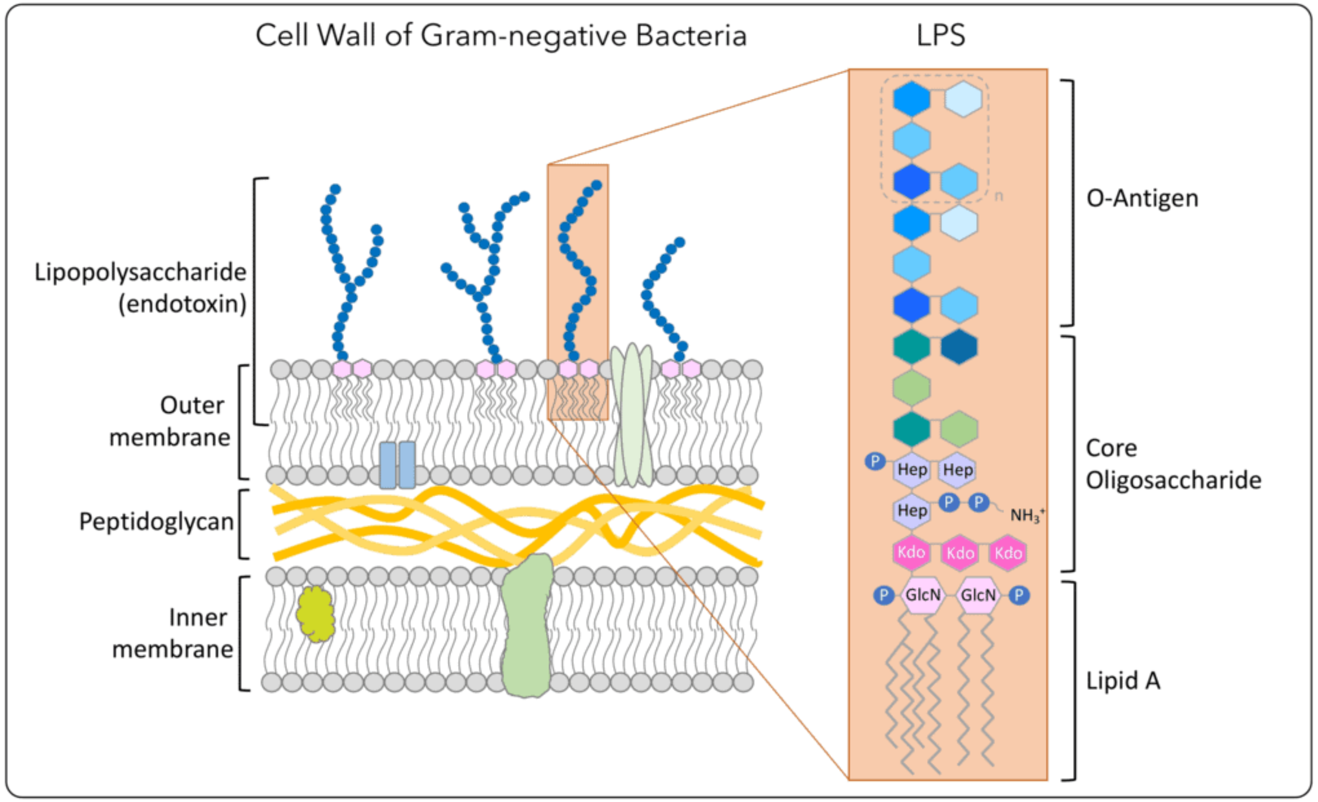
Endotoxins



- Lipopolysaccharide (LPS) complex
- Components of the gram-negative outer cell membrane
- Most potent pyrogens

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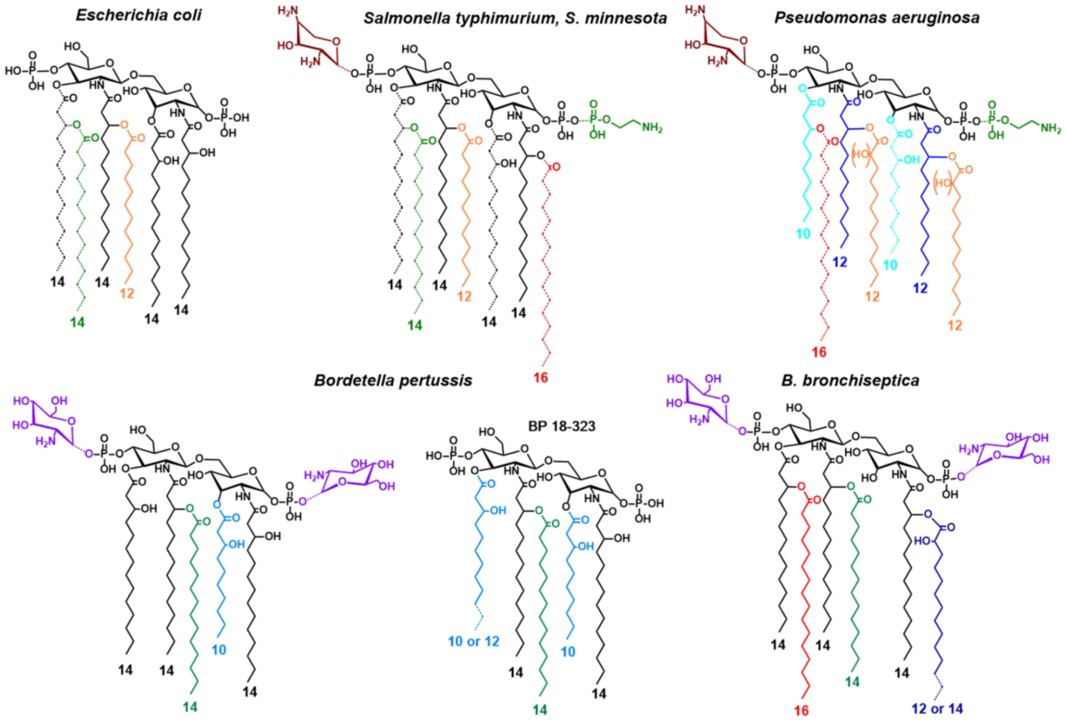
Endotoxins

Lipid A structural diversity

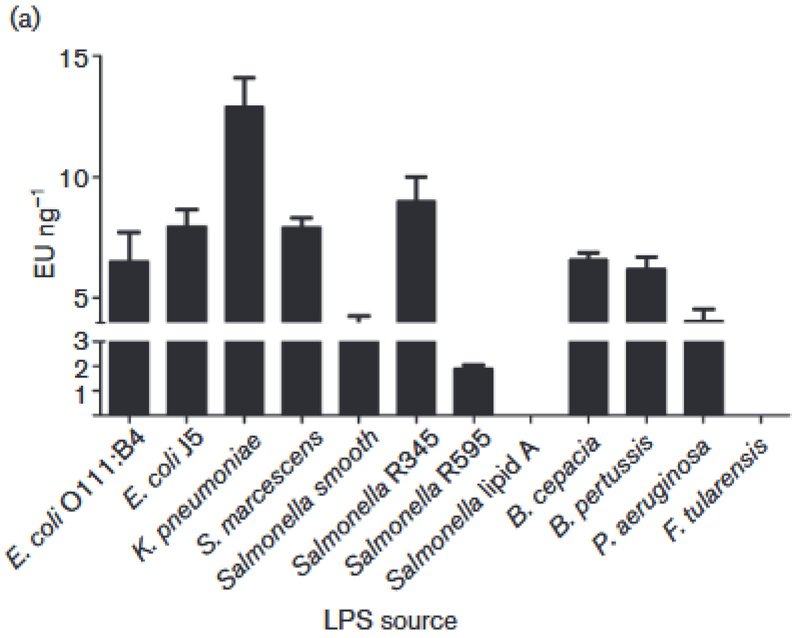


Caroff and Novikov (2020)

- Lipopolysaccharide (LPS) complex
- Components of the gram-negative outer cell membrane
- Most potent pyrogens

Endotoxins

Reactivity of LPS from various organisms

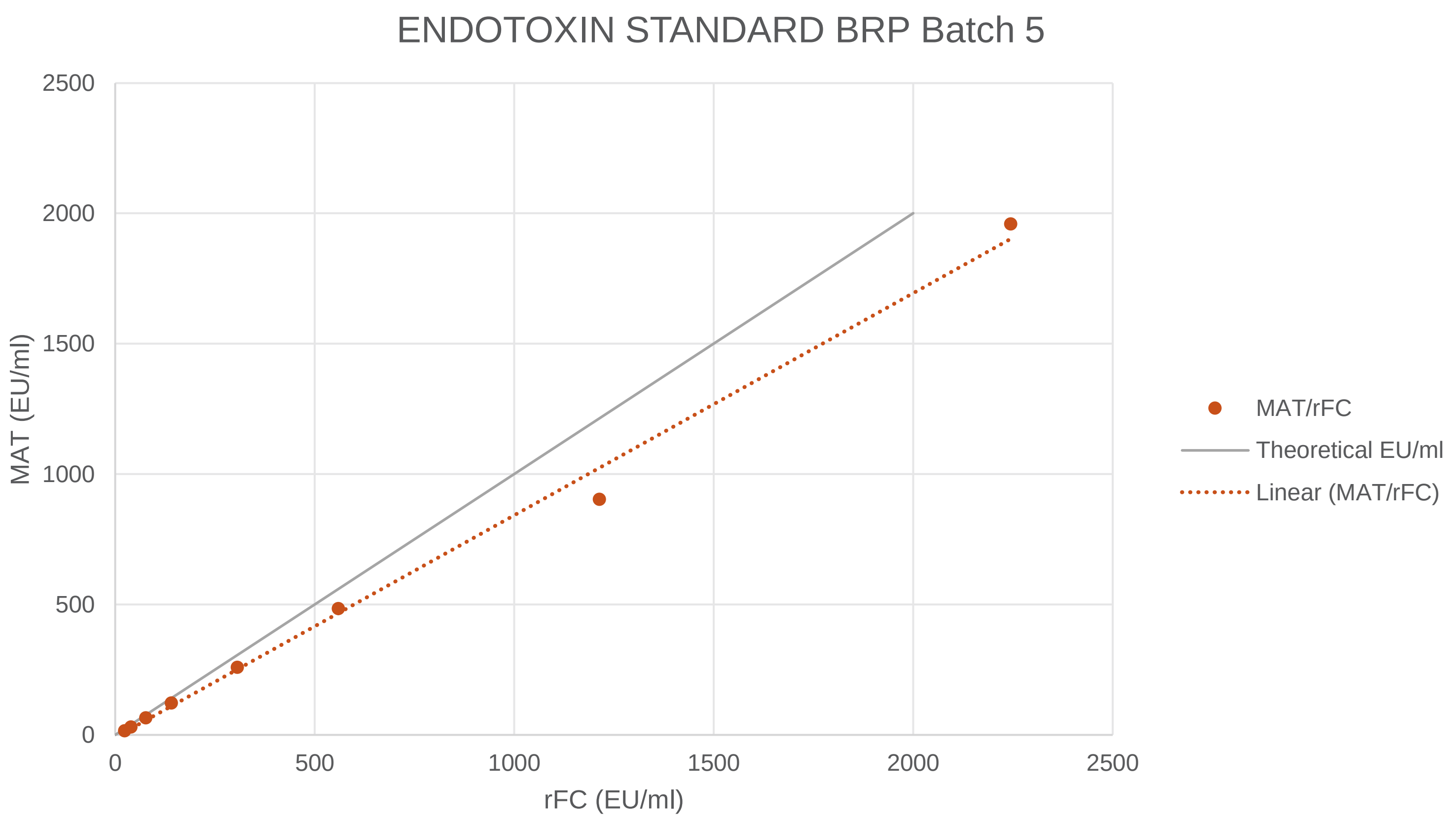


Abate et al. (2017)

- Lipopolysaccharide (LPS) complex
- Components of the gram-negative outer cell membrane
- Most potent pyrogens

MAT vs. rFC: detection of endotoxin standard

Theoretical value EDQM standard (EU/ml)	rFC (EU/ml)	MAT (EEU/ml)
15,625	23,2	15,9
31,25	39,4	30
62,5	76,4	65,7
125	140,6	122,3
250	306,2	258,5
500	559,3	483,6
1000	1213,6	903,1
2000	2244,5	1959,7



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